

# CURRICULUM VITAE

Filippo Bistaffa

July 2026

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## Personal Information

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- **Email:** [filippo.bistaffa@gmail.com](mailto:filippo.bistaffa@gmail.com)
- **Webpage:** <https://filippobistaffa.github.io>
- **ORCID:** 0000-0003-1658-6125
- **Web of Science ID:** AAA-4942-2020
- **SCOPUS ID:** 55516740200

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## Academic Positions

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| Since 12/2023     | <b>Tenured Researcher</b><br>Department of Multi-Agent Systems, IIIA-CSIC, Barcelona, Spain                |
| 08/2019 – 11/2023 | <b>Postdoctoral Research Fellow</b><br>Department of Multi-Agent Systems, IIIA-CSIC, Barcelona, Spain      |
| 06/2017 – 06/2019 | <b>Marie-Curie Fellow</b><br>Department of Multi-Agent Systems, IIIA-CSIC, Barcelona, Spain                |
| 01/2016 – 06/2017 | <b>Postdoctoral Research Fellow</b><br>Department of Computer Science, University of Verona, Verona, Italy |

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## Biography

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During my career, I have been focusing on the **multi-perspective study of Constrained Optimisation**, making **strong contributions to the state-of-the-art** both on **theoretical** and **practical** aspects and by tackling different **real-world domains**, such as collective energy purchasing, sustainable mobility, cooperative learning, high-performance computing, and AI-human value alignment. I received my PhD (Doctor Europæus) in April 2016 at the University of Verona with a thesis entitled “Constraint Optimisation Techniques for Real-World Applications”, which was given the “Young Doctors Award” (Special Mention) by the Italian Association for AI (AIxIA). In 2017, I was awarded a **Marie Skłodowska-Curie Individual Fellowship** (the most prestigious post-doctoral fellowship at the European level), which I spent at the IIIA Institute of the Spanish National Research Council (CSIC). I led (as the **Scientific Principal Investigator**) the AVI-IA project (privately funded by the ITBID S.L. company), which aimed to implement an AI system that guides strategic decisions in purchasing processes by exploiting novel Constrained Optimisation techniques that I developed. The resulting AI algorithms have been fully integrated into ITBID’s commercial product with a Technology Readiness Level of 7 (<https://bit.ly/avi-ia>). I am currently a **CSIC Tenured Researcher** (*Científico Titular*) and I am currently leading, as the **Co-Principal Investigator** in collaboration with the CSIC’s Unity of Institutional Intelligence (UNIN), a MOMENTUM project aimed at devising AI solutions for the CSIC’s *Institutional Strategy*, i.e., to improve the efficiency of the data management and processes of the CSIC’s administration through AI.

Indicators:     **Google Scholar:** Citations: 877, h-index: 17, i10-index: 22  
                  **Scopus:** Citations: 470, h-index: 14

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## Graduate Studies

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| 04/2016 | <b>Ph.D. (Doctor Europæus) Computer Science</b> , University of Verona<br><b>Thesis:</b> Constraint Optimisation Techniques for Real-World Applications<br><b>Supervisor:</b> Prof. Alessandro Farinelli<br><b>Special mention:</b> Italian Association for AI (AIxIA) “Young Doctors Award” 2017 |
| 03/2012 | <b>M.Sc. Engineering and Computer Science</b> , University of Verona<br><b>Thesis:</b> Coalition Formation in the Energy Market with a Graphical Model Approach<br><b>Supervisor:</b> Prof. Alessandro Farinelli                                                                                  |
| 10/2009 | <b>B.Eng. Computer Engineering</b> , University of Pavia<br><b>Thesis:</b> Implementation of Network Lossy Links with Kernel Linux 2.6<br><b>Supervisor:</b> Prof. Giuseppe Rossi                                                                                                                 |

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## Participation in Research Projects

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| 06/2024 – 02/2026 | <b>Title:</b> Ethically Aligned Robot for the Care of Autistic Children (EMROBCARE)<br><b>Position:</b> Tenured Researcher, IIIA-CSIC, Barcelona, Spain<br><b>Funding:</b> SEDIA (907,169 euro)                                                  |
| 01/2025 – 12/2028 | <b>Title:</b> AI for CSIC’s Institutional Strategy and Data Management (AI-UNIN)<br><b>Position:</b> <u>Co-Principal Investigator</u> , IIIA-CSIC, Barcelona, Spain<br><b>Funding:</b> Momentum/MTDFP (293,933 euro)                             |
| 10/2023 – 08/2026 | <b>Title:</b> Advanced Computing Techniques to Achieve Sustainable Development (ACISUD)<br><b>Position:</b> Tenured Researcher, IIIA-CSIC, Barcelona, Spain<br><b>Funding:</b> MICINN (183,500 euro)                                             |
| 12/2022 – 05/2024 | <b>Title:</b> Sustainable Urban Transportation and Delivery of Goods (E-TRANS)<br><b>Position:</b> Tenured Researcher, IIIA-CSIC, Barcelona, Spain<br><b>Funding:</b> Spanish State Research Agency, AEI (91,425 euro)                           |
| 11/2022 – 05/2024 | <b>Title:</b> Youth Marketplace Operational Research (YOMA OR)<br><b>Position:</b> Tenured Researcher, IIIA-CSIC, Barcelona, Spain<br><b>Funding:</b> Fondation Botnar (800,000 euro)                                                            |
| 10/2021 – 10/2022 | <b>Title:</b> An AI-Based System to Support Purchasing Processes (AVI-IA)<br><b>Position:</b> <u>Scientific Principal Investigator</u> , IIIA-CSIC, Barcelona, Spain<br><b>Funding:</b> Private Contract with ITBID Company (119,500 euro)       |
| 11/2020 – 10/2022 | <b>Title:</b> High-Performance Optimisation for Artificial Collective Intelligence (HIMPACT)<br><b>Position:</b> <u>Principal Investigator</u> , IIIA-CSIC, Barcelona, Spain<br><b>Funding:</b> Spanish State Research Agency, AEI (10,000 euro) |
| 06/2020 – 05/2023 | <b>Title:</b> Computational Intelligence for Sustainable Development Goals (CI-SUSTAIN)<br><b>Position:</b> Postdoctoral Research Fellow, IIIA-CSIC, Barcelona, Spain<br><b>Funding:</b> MICINN (102,850 euro)                                   |
| 08/2019 – 09/2021 | <b>Title:</b> Data Management Techniques for Real-Time Logistics Planning (LOGISTAR)<br><b>Position:</b> Postdoctoral Research Fellow, IIIA-CSIC, Barcelona, Spain<br><b>Funding:</b> H2020 (5,000,000 euro)                                     |
| 06/2017 – 06/2019 | <b>Title:</b> Collectiveware: Highly-parallel Algorithms for Collective Intelligence (HPA4CF)<br><b>Position:</b> <u>Marie-Curie Fellow</u> , IIIA-CSIC, Barcelona, Spain<br><b>Funding:</b> H2020-MSCA-IF-2016 (158,121 euro)                   |
| 01/2016 – 06/2017 | <b>Title:</b> Development of AI Techniques for Sustainable Commuting (AI-SC)<br><b>Position:</b> Postdoctoral Research Fellow, University of Verona, Italy<br><b>Funding:</b> University of Verona (29,000 euro)                                 |

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## Fellowships & Awards

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- **Marie Skłodowska-Curie Individual Fellowship** (H2020-MSCA-IF-2016)  
13% funded proposals out of 832 under “Information Science and Engineering” descriptor
- **Juan de la Cierva Formación** (FJC-2016, renounced due to incompatibility with MSCA fellowship)  
1<sup>st</sup> place in “Computation and Information Technology Sciences” ranking

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## Supervised Students

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|              |                                                                                                                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021 – 2026  | Roger Xavier Lera Leri ( <b>Ph.D.</b> )<br>Thesis: Explainability for Optimisation-based Decision Support Systems<br>Department of Multi-Agent Systems, IIIA-CSIC, Barcelona, Spain<br>Graduation Date: 16/06/2026                                                        |
| 2021 – 2025  | Camilo Chacón Sartori ( <b>Ph.D.</b> , co-supervised with Dr. Christian Blum)<br>Thesis: Improving Optimization Algorithms via Machine Learning and Visualization Tools<br>Department of Multi-Agent Systems, IIIA-CSIC, Barcelona, Spain<br>Graduation Date: 12/12/2025  |
| 2019 – 2023  | Adrià Fenoy ( <b>Ph.D.</b> , co-supervised with Prof. Alessandro Farinelli)<br>Thesis: Combining Optimisation and Machine Learning for the Formation of Collectives<br>Department of Computer Science, University of Verona, Verona, Italy<br>Graduation Date: 28/08/2023 |
| <b>M.Sc.</b> | 9 students (2025, 2 in 2024, 2022, 2021, 2020, 2 in 2019, 2015)                                                                                                                                                                                                           |

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## Technology Contributions

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- **Notarial Registry:** “Bachelor’s Degree Planning”: Personalized Education for Skills’ Acquisition  
**Reference:** N° 288/2025, (registered by CSIC on the 12<sup>th</sup> of December 2025)
  - **Notarial Registry:** “Learning Path Planner”: Personalised Education Paths for Students  
**Reference:** N° 369/2024, (registered by CSIC on the 17<sup>th</sup> of December 2024)
  - **Notarial Registry:** “Green Routes”: A Software to Compute Green Routes for Pedestrians  
**Reference:** N° 171/2024, (registered by CSIC on the 15<sup>th</sup> of July 2024)
  - **Notarial Registry:** “SynTeam”: Team Formation Optimisation for Cooperative Learning  
**Reference:** N° 2301/2020, (registered by CSIC on the 29<sup>th</sup> of June 2020)
  - **Notarial Registry:** “CoMe4Ride”: Large-Scale Online Ridesharing Optimisation Software  
**Reference:** N° 3757/2019, (registered by CSIC on the 18<sup>th</sup> of October 2019)
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## Teaching Experience

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|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 – 2025   | <b>Lecturer</b> – Planning Approaches for Autonomous Agents and Multi-Agent Systems<br>Master’s Degree in Artificial Intelligence, University of Verona, Verona, Italy (~50 students) |
| Since 2022    | <b>Lecturer</b> – Tutorial on Multi-Agent Distributed Constrained Optimization<br>AAMAS 2026 (Cyprus), 2024 (New Zealand), 2023 (UK), 2022 (Online) (~40 attendees)                   |
| 10/06/2025    | <b>Lecturer</b> – Optimisation in Multi-Agent Systems<br>AI4ALL Training Program, Online (~25 attendees)                                                                              |
| 4–9/07/2022   | <b>Lecturer</b> – Computational Sustainability in Multi-Agent Systems<br>AI-HUB Summer School, Palma de Mallorca, Spain (95 attendees)                                                |
| 19–23/07/2021 | <b>Lecturer</b> – Constrained Optimisation for Multi-Agent Systems<br>European Agent Systems Summer School (EASSS), Online (110 attendees)                                            |
| 1–5/07/2019   | <b>Lecturer</b> – Constrained Optimisation for Multi-Agent Systems<br>Advanced Course on AI (ACAI) Summer School, Crete, Greece (90 attendees)                                        |
| 2013 – 2016   | <b>Tutor</b> – Algorithms (Programming Laboratory II)<br>University of Verona, Verona, Italy (~40 students/year)                                                                      |

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## Journal Publications\*

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- [1] Manel Rodriguez-Soto, Roxana Rădulescu, **Filippo Bistaffa**, Oriol Ricart, Arnau Mayoral, Maite Lopez-Sanchez, Juan A. Rodriguez-Aguilar, Ann Nowé. Multi-Objective Reinforcement Learning for Provably Incentivising Alignment with Value Systems. In *Artificial Intelligence*, 2026 (Q2, **D1**, **D1**). DOI: 10.1016/j.artint.2025.104460.
- [2] Roger Lera-Leri, **Filippo Bistaffa**, Tomas Trescak, Juan A. Rodriguez-Aguilar. Computing Job-Tailored Degree Plans towards the Acquisition of Professional Skills. In *Annals of Operations Research*, 2025 (**Q1**, **Q1**, **Q1**). DOI: 10.1007/s10479-025-06678-6.
- [3] Camilo Chacón Sartori, Christian Blum, **Filippo Bistaffa**. VisGraphVar: A Benchmark Generator for Assessing Variability in Graph Analysis Using Large Vision-Language Models. In *IEEE Access*, 2025 (Q2, **D1**, Q1). DOI: 10.1109/ACCESS.2025.3535837.
- [4] Camilo Chacón Sartori, Christian Blum, **Filippo Bistaffa**, Guillem Rodríguez Corominas. Metaheuristics and Large Language Models Join Forces: Towards an Integrated Optimization Approach. In *IEEE Access*, 2025 (Q2, **D1**, Q1). DOI: 10.1109/ACCESS.2024.3524176.
- [5] Sergio Calo, **Filippo Bistaffa**, Anders Jonsson, Vicenç Gomez, Mar Viana. Spatial Air Quality Prediction in Urban Areas via Message Passing. In *Engineering Applications of Artificial Intelligence*, 2024 (**D1**, Q1, **D1**). DOI: 10.1016/j.engappai.2024.108191.
- [6] Roger Lera-Leri, Enrico Liscio, **Filippo Bistaffa**, Catholijn Jonker, Maite Lopez-Sanchez, Juan A. Rodríguez-Aguilar, Pradeep Murukannaiah, Francisco Salas-Molina. Aggregating Value Systems for Decision Support. In *Knowledge-Based Systems*, 2024 (Q1, **D1**, **D1**). DOI: 10.1016/j.knosys.2024.111453.
- [7] Adrià Fenoy, **Filippo Bistaffa**, Alessandro Farinelli. An Attention Model for the Formation of Collectives in Real-World Domains. In *Artificial Intelligence*, 2024 (Q2, **D1**, **D1**). DOI: 10.1016/j.artint.2023.104064.
- [8] Marc Serramia, Manel Rodriguez-Soto, Maite Lopez-Sanchez, Juan A. Rodríguez-Aguilar, **Filippo Bistaffa**, Paula Boddington, Michael Wooldridge, Carlos Ansoategui. Encoding Ethics to Compute Value-Aligned Norms. In *Minds & Machines*, 2023 (Q2, **D1**, **D1**). DOI: 10.1007/s11023-023-09649-7.
- [9] Francisco Salas-Molina, **Filippo Bistaffa**, Juan A. Rodríguez-Aguilar. A General Approach for Computing a Consensus in Group Decision Making that Integrates Multiple Ethical Principles. In *Socio-Economic Planning Sciences*, 2023 (**D1**, **D1**, **D1**). DOI: 10.1016/j.seps.2023.101694.
- [10] Dave de Jonge, **Filippo Bistaffa**, Jordi Levy. Multi-Objective Vehicle Routing with Automated Negotiation. In *Applied Intelligence*, 2022 (Q2, Q2, Q2). DOI: 10.1007/s10489-022-03329-2.
- [11] **Filippo Bistaffa**, Georgios Chalkiadakis, Alessandro Farinelli. Efficient Coalition Structure Generation via Approximately Equivalent Induced Subgraph Games. In *IEEE Transactions on Cybernetics*, 2022 (**D1**, **D1**, **D1**). DOI: 10.1109/TCYB.2020.3040622.
- [12] **Filippo Bistaffa**, Christian Blum, Jesús Cerquides, Alessandro Farinelli, Juan A. Rodríguez-Aguilar. A Computational Approach to Quantify the Benefits of Ridesharing for Policy Makers and Travellers. In *IEEE Transactions on Intelligent Transportation Systems*, 2021 (**D1**, **D1**, **D1**). DOI: 10.1109/TITS.2019.2954982.
- [13] Ewa Andrejczuk, **Filippo Bistaffa**, Christian Blum, Juan A. Rodríguez-Aguilar, Carles Sierra. Synergistic Team Composition: A Computational Approach to Foster Diversity in Teams. In *Knowledge-Based Systems*, 2019 (Q1, **D1**, **D1**). DOI: 10.1016/j.knosys.2019.06.007.
- [14] Michele Roncalli, **Filippo Bistaffa**, Alessandro Farinelli. Decentralized Power Distribution in the Smart Grid with Ancillary Lines. In *Mobile Networks and Applications*, 2019 (Q2, Q1, Q2). DOI: 10.1007/s11036-017-0893-y.
- [15] **Filippo Bistaffa** and Alessandro Farinelli. A COP Model for Graph-Constrained Coalition Formation. In *Journal of Artificial Intelligence Research*, 2018 (Q3, Q2, Q1). DOI: 10.1613/jair.1.11205.

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\*Each article reports the quantiles published by JCR, Scopus, and Scimago in the year of publication or the most recent available, highlighting the highest one (source: GesBIB). Following the indications by GesBIB (slide 5), the reported quantile for each provider is the maximum across all the categories of the journal.

- [16] **Filippo Bistaffa**, Georgios Chalkiadakis, Alessandro Farinelli, Sarvapali D. Ramchurn. A Cooperative Game-Theoretic Approach to the Social Ridesharing Problem. In *Artificial Intelligence*, 2017 (Q1, **D1**, **D1**). DOI: 10.1016/j.artint.2017.02.004.
- [17] **Filippo Bistaffa**, Alessandro Farinelli, Jesús Cerquides, Juan A. Rodríguez-Aguilar, Sarvapali D. Ramchurn. Algorithms for Graph-Constrained Coalition Formation in the Real World. In *ACM Transactions on Intelligent Systems and Technology*, 2017 (Q1, **D1**, Q1). DOI: 10.1145/3040967.
- [18] **Filippo Bistaffa**, Nicola Bombieri, Alessandro Farinelli. An Efficient Approach for Accelerating Bucket Elimination on GPUs. In *IEEE Transactions on Cybernetics*, 2017 (**D1**, **D1**, **D1**). DOI: 10.1109/TCYB.2016.2593773.
- [19] Alessandro Farinelli, Manuele Bicego, **Filippo Bistaffa**, Sarvapali D. Ramchurn. A Hierarchical Clustering Approach to Large-Scale Near-Optimal Coalition Formation with Quality Guarantees. In *Engineering Applications of Artificial Intelligence*, 2017 (**Q1**, **Q1**, **Q1**). DOI: 10.1016/j.engappai.2016.12.018.

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## International Conference Publications

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- [20] Roger Lera-Leri, **Filippo Bistaffa**, Juan A. Rodríguez-Aguilar. Predicting the Cost of Explanations in Optimisation Problems. In *Lecture Notes in Computer Science* (LNAI), 2026.
- [21] Dimitrios Troullinos, María Nuria Conejero, **Filippo Bistaffa**, José María Bengochea-Guevara, Ángela Ribeiro, Juan A. Rodríguez-Aguilar. RoboVineSim: A Simulation Tool for Human-Robot Collaboration in Vineyard Harvesting (Demonstration Track). In *International Joint Conference on Artificial Intelligence* (IJCAI), 2026 (CORE **A\***).
- [22] Roger Lera-Leri, **Filippo Bistaffa**, Athina Georgara, Juan A. Rodríguez-Aguilar. Exploiting Constraint Reasoning to Build Graphical Explanations for Mixed-Integer Linear Programming. In *Lecture Notes in Computer Science* (LNAI), 2025. DOI: 10.1007/978-3-032-01399-6\_2.
- [23] **Filippo Bistaffa**, Sergio Calo. Recommending Green Routes for Pedestrians to Reduce the Exposure to Air Pollutants in Barcelona (Demonstration Track). In *International Conference on Autonomous Agents and Multiagent Systems* (AAMAS), 2025 (CORE **A\***). DOI: 10.5555/3709347.3744077.
- [24] Adrià Fenoy, **Filippo Bistaffa**, Alessandro Farinelli. An Attention Model for the Formation of Collectives in Real-World Domains. **Invited** in *European Conference of Artificial Intelligence* (ECAI), 2024 (CORE **A**). DOI: 10.3233/FAIA240468.
- [25] Roger Lera-Leri, **Filippo Bistaffa**. A Robust and Scalable Approach to Meet User Preferences in Research Project Planning. In *European Conference of Artificial Intelligence* (ECAI), 2024 (CORE **A**). DOI: 10.3233/FAIA241043.
- [26] Adrià Fenoy, Jacopo Zagoli, **Filippo Bistaffa**, Alessandro Farinelli. Attention for the Allocation of Tasks in Multi-Agent Pickup and Delivery. In *ACM Symposium on Applied Computing* (ACM SAC), 2024 (CORE **B**). DOI: 10.1145/3605098.3635955.
- [27] Errikos Streviniotis, Athina Georgara, **Filippo Bistaffa**, Georgios Chalkiadakis. FairPlay: A Multi-Sided Fair Dynamic Pricing Policy for Hotels. In *AAAI Conference on Artificial Intelligence* (AAAI), 2024 (CORE **A\***). DOI: 10.1609/aaai.v38i20.30243.
- [28] **Filippo Bistaffa**. Faster Exact MPE and Constrained Optimization with Deterministic Finite State Automata. In *International Joint Conference on Artificial Intelligence* (IJCAI), 2023 (CORE **A\***). DOI: 10.24963/ijcai.2023/209.
- [29] Enrico Liscio, Roger Lera-Leri, **Filippo Bistaffa**, Roel Dobbe, Catholijn Jonker, Maite Lopez-Sanchez, Juan A. Rodríguez-Aguilar, Pradeep Murukannaiah. Value Inference in Sociotechnical Systems. In *International Conference on Autonomous Agents and Multiagent Systems* (AAMAS), 2023 (CORE **A\***). DOI: 10.5555/3545946.3598838.
- [30] Tomas Trescak, Roger Lera Leri, **Filippo Bistaffa**, Juan A. Rodríguez-Aguilar. Agent-Assisted Life-Long Education and Learning. In *International Conference on Autonomous Agents and Multiagent Systems* (AAMAS), 2022 (CORE **A\***). DOI: 10.5555/3535850.3536117.
- [31] Roger Lera-Leri, **Filippo Bistaffa**, Marc Serramia, Maite Lopez-Sanchez, Juan A. Rodríguez-Aguilar. Towards Pluralistic Value Alignment: Aggregating Value Systems through  $\ell_p$ -Regression. In *International Conference on Autonomous Agents and Multiagent Systems* (AAMAS), 2022 (CORE **A\***). DOI: 10.5555/3535850.3535938.

- [32] Dave de Jonge, **Filippo Bistaffa**, Jordi Levy. A Heuristic Algorithm for Multi-Agent Vehicle Routing with Automated Negotiation. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2021 (CORE A\*). DOI: 10.5555/3463952.3464004.
- [33] Ewa Andrejczuk, **Filippo Bistaffa**, Christian Blum, Juan A. Rodríguez-Aguilar, Carles Sierra. Heterogeneous Teams for Homogeneous Performance. In *Conference on Principles and Practice of Multiagent Systems (PRIMA)*, 2018 (CORE B). DOI: 10.1007/978-3-030-03098-8\_6.
- [34] **Filippo Bistaffa** and Alessandro Farinelli. A COP Model for Graph-Constrained Coalition Formation. Invited in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2018 (CORE A\*). DOI: 10.24963/ijcai.2018/783.
- [35] Ewa Andrejczuk, **Filippo Bistaffa**, Christian Blum, Juan A. Rodríguez-Aguilar, Carles Sierra. Solving the Synergistic Team Composition Problem. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2018 (CORE A\*). DOI: 10.5555/3237383.3238001.
- [36] **Filippo Bistaffa**, Juan A. Rodríguez-Aguilar, Jesús Cerquides, Christian Blum. A Simulation Tool for Large-Scale Online Ridesharing (Demonstration). In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2018 (CORE A\*). DOI: 10.5555/3237383.3237981.
- [37] **Filippo Bistaffa**, Nicola Bombieri, Alessandro Farinelli. CUBE: A CUDA Approach for Bucket Elimination on GPUs. In *European Conference on Artificial Intelligence (ECAI)*, 2016 (CORE A). DOI: 10.3233/978-1-61499-672-9-125.
- [38] **Filippo Bistaffa**, Georgios Chalkiadakis, Alessandro Farinelli, Sarvapali D. Ramchurn. Recommending Fair Payments for Large-Scale Social Ridesharing. In *ACM Conference on Recommender Systems (RecSys)*, 2015 (CORE A). DOI: 10.1145/2792838.2800177.
- [39] **Filippo Bistaffa**, Alessandro Farinelli, Sarvapali D. Ramchurn. Sharing Rides with Friends: a Coalition Formation Algorithm for Ridesharing. In *AAAI Conference on Artificial Intelligence (AAAI)*, 2015 (CORE A\*). URL: <http://www.aaai.org/ocs/index.php/AAAI/AAAI15/paper/view/9622>.
- [40] **Filippo Bistaffa**, Alessandro Farinelli, Nicola Bombieri. Optimising Memory Management for Belief Propagation in Junction Trees Using GPGPUs. In *International Conference on Parallel and Distributed Systems (ICPADS)*, 2014 (CORE B). DOI: 10.1109/padsw.2014.7097850.
- [41] **Filippo Bistaffa**, Alessandro Farinelli, Jesús Cerquides, Juan A. Rodríguez-Aguilar, Sarvapali D. Ramchurn. Anytime Coalition Structure Generation on Synergy Graphs. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2014 (CORE A\*). DOI: 10.5555/2615731.2615737.
- [42] **Filippo Bistaffa**. Parallel Algorithms for Hard Combinatorial Optimisation Problems in Multi-Agent Systems (Doctoral Consortium). In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2014 (CORE A\*). DOI: 10.5555/2615731.2616142.
- [43] **Filippo Bistaffa** and Alessandro Farinelli. A Fast Approach to Form Core-Stable Coalitions Based on a Dynamic Model. In *IEEE/WIC/ACM International Joint Conferences on Web Intelligence and Intelligent Agent Technologies (WI/IAT)*, 2013 (CORE B). DOI: 10.1109/WI-IAT.2013.100.
- [44] **Filippo Bistaffa**, Alessandro Farinelli, Meritxell Vinyals, Alex Rogers. Coalitional Energy Purchasing in the Smart Grid. In *IEEE International Energy Conference & Exhibition (ENERGYCON)*, 2012. DOI: 10.1109/energycon.2012.6348270.
- [45] **Filippo Bistaffa**, Alessandro Farinelli, Meritxell Vinyals, Alex Rogers. Decentralised Stable Coalition Formation Among Energy Consumers in the Smart Grid (Demonstration). In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2012 (CORE A\*). DOI: 10.5555/2343896.2344061.

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## Scientific Community Service

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- Organiser and Chair of OptLearnMAS workshop at AAMAS conference, since 2023
- IIIA coordinator of CSIC's Interdisciplinary Platform on Mobility (PTI Mobility 2030), since 2020
- Member of the Evaluation Committee for HORIZON-MSCA-2022-COFUND program
- Reviewer for journals: AIJ, IEEE TCYB, EJOR, IEEE T-ITS, JAIR, JAAMAS
- Area Chair of the Search, Optimization, Planning, and Scheduling (SOPS) track at AAMAS, since 2026
- Senior Program Committee member for conferences: IJCAI, AAMAS, ECAI
- Program Committee member for conferences: AAAI, GECCO, EuMAS, ACM SAC
- Board member for ACIA (Associació Catalana d'Intel·ligència Artificial) M.Sc. thesis award, 2019

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## Selected Dissemination & Outreach Activities

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| 2026       | How to Avoid Urban Collapse? Towards a Sustainable Urban Mobility Model (Collaborator)<br>Book part of the <i>Science4Policies</i> series published by the CSIC                                           |
| 2024       | IA: Transforming Policies and Public Goods Management in the Digital Era (Collaborator)<br>Book part of the <i>Science4Policies</i> series published by the CSIC                                          |
| 16/10/2024 | AI, Sustainability and New Technology for Cultural Events (Invited Panel Member)<br>Autonomous University of Barcelona, Spain ( <a href="https://youtu.be/3Zcptxnz5Bc">https://youtu.be/3Zcptxnz5Bc</a> ) |
| 13/07/2023 | AI Techniques for Sustainable Shared Mobility (Barcelona Metropolis, Barcelona City Hall)<br>Awarded by the ACIA the 2 <sup>nd</sup> Prize for the Best Artificial Intelligence Dissemination Work        |
| 08/07/2022 | Mathematical Education & Artificial Intelligence (Moderator, AI-HUB Summer School)<br>CaixaForum, Palma de Mallorca, Spain ( <a href="https://youtu.be/V1BjRm2XFZ8">https://youtu.be/V1BjRm2XFZ8</a> )    |
| 09/12/2021 | AI for Shared Mobility (Invited Seminar by Zaragoza's Chamber of Commerce)<br>University of Zaragoza, Zaragoza, Spain                                                                                     |
| 24/03/2021 | AI for Shared Mobility (Invited Seminar by IberCaja Foundation)<br>Online, <a href="https://youtu.be/moS77BKEYE">https://youtu.be/moS77BKEYE</a>                                                          |
| 03/12/2020 | Large-Scale P2P Ridesharing (Invited Seminar)<br>University of Girona, Girona, Spain                                                                                                                      |
| 28/02/2020 | Presentation of PTI Mobility 2030 (Invited Talk)<br>ICB-CSIC, Zaragoza, Spain ( <a href="https://youtu.be/9fii0vcmxwA">https://youtu.be/9fii0vcmxwA</a> )                                                 |

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## International Research Visits

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|-------------------|----------------------------------------------------------------|
| 06/2014 – 03/2015 | ECS Department, University of Southampton, United Kingdom      |
| 06/2013           | Department of Multi-Agent Systems, IIIA-CSIC, Barcelona, Spain |